

JEE MAIN REVISION MASTER: DUAL NATURE OF RADIATION SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 207 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Dual Nature Of Radiation - Core Application	207 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dual Nature Of Radiation - Core Application (Very High Yield — 207 PYQs)

Core Principle: Key problem pattern in Dual Nature Of Radiation.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons /s . The region of an electromagnetic spectrum to wh..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Light source having wavelength 331 nm is used to generate photo-electrons whose stopping potential is 0.2 V . The work function of the used ..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is _____. (Given : $h=6.63 \times 10^{-34}$..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons /s . The region of an electromagnetic spectrum to which the emitted electromagnetic radiation belongs to ____ (Take $h=6.6 \times 10^{-34}\text{J.s}$ and $c=3 \times 10^8\text{m/s}$).

- (A) Microwave
- (B) Infrared
- (C) Visible
- (D) Ultraviolet [Correct]

Correct Answer: D

Solution Analysis: The power (P) of a light source is the total energy emitted per unit time. If the source emits n photons per second, and each photon has an energy E , then the power is: $P=n \times E$ The given power of the source is $P=15\text{kW}=15 \times 10^3\text{W}$ and the number of photons per second is $n=2.5 \times 10^{22}\text{s}^{-1}$ $E=(P / n) \Rightarrow E=\frac{15 \times 10^3}{2.5 \times 10^{22}} \Rightarrow E=6 \times 10^{-19}\text{J}$ The energy of a photon is related to its wavelength by the formula: $E=(hc / \lambda) \Rightarrow \lambda=(hc / E)$ Substituting the constants and energy, The PI